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International Journal of Epidemiology 1998;27:231–237

Night blindness of pregnancy in rural Nepal—nutritional and health risks

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Background Night blindness (XN) is the most common clinical symptom of vitamin A deficiency among children in developing countries. Yet little is known about the aetiology or associated risks of maternal XN. Emerging evidence from South East Asia suggests that it may be more frequent than previously thought in women of reproductive age, especially during pregnancy.

Methods A population-based, case-control study was conducted to reveal the epidemiology of XN among pregnant Nepali women. Night blind cases were identified by history through a weekly community surveillance system. Controls were randomly selected from a pool of pregnant women without XN and pair-matched for gestational age of the cases. A home-based assessment was done within a week of selection, at which 7-day food frequency and morbidity histories were collected, anthropometry measured, and capillary blood drawn for serum retinol, β-carotene and haemoglobin (Hb) estimation.

Results Cases and controls did not differ by age or number of previous pregnancies. However, cases were more likely to be from the lower castes, be illiterate, live in poorer quality homes, and own no land. The mean serum retinol level of cases was ~0.30 μmol/l lower than controls ($P < 0.001$), indicating a low vitamin A status of night blind pregnant women. Mean Hb level was significantly lower (by 0.7 g/dl, $P < 0.004$), and the risk of severe anaemia (Hb <7.0 g/dl) higher among cases than controls (odds ratio = 3.0, 95% CI : 1.25–7.23). Cases were more undernourished than controls reflected by lower mean weight (–2.6 kg), body mass index (–0.8), arm circumference (–0.9 cm) and triceps skinfold (–0.8 mm). Night blindness was associated with less frequent consumption of preformed vitamin A (milk products, fish and meat) and provitamin A (dark green leafy vegetables and mangoes) foods, especially in summer. Night blind women were 2–3 times more likely to report symptoms of urinary/reproductive tract infections such as lower abdominal pain, painful and burning urination, or vaginal discharge, symptoms of diarrhoea/dysentery, of pre-eclampsia or eclampsia, and of nausea, vomiting or poor appetite throughout pregnancy than controls.

Conclusion Women who experience XN during pregnancy have a low vitamin A status, although several other risk factors appear to cluster among these women as well. Night blind women are also more likely to be anaemic, ill, and acutely undernourished, and to be consuming a nutritionally poorer diet in pregnancy than non-night blind pregnant women. A simple history of XN can identify women at high risk during pregnancy who may require special nutritional support, antenatal care and counselling.

Keywords Night blindness, pregnancy, vitamin A deficiency, Nepal

Accepted 7 July 1997

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Maternal night blindness (XN) has received little attention as a problem of potential public health significance in developing countries. Early accounts, found in two clinic-based studies in the late 1960s from India, reported that XN occurred frequently among pregnant Indian women, most often during the third trimester followed by spontaneous remission soon after parturition.[1, 2] Some amelioration following treatment with high doses of vitamin A was observed among pregnant women who presented at the clinic with XN. Recent studies suggest XN to be widespread, affecting 10–50% of women during pregnancy in rural South East Asia.[3–6] Night blindness in these studies was ascertained by obtaining a simple history from the women, either concurrently or retrospectively. Although maternal XN is believed to be a manifestation of vitamin A deficiency, as it is in young children, none of these studies have examined this association. In the south-eastern plains of Nepal, XN affects 10–20% of women during pregnancy, tends to recur with repeated pregnancies[3] and is recognized as an important health problem among women.[7] Night blindness in pregnancy, thus, exists among South East Asian women but its aetiology or associated risk factors, severity, consequences, and public health significance have not been elucidated.

We report findings of a population based case-control study that examined nutritional and other risk factors associated with XN during pregnancy in rural Nepal.

Methods

Study area and population

The study was conducted in Sarlahi District, in the low-lying, south-eastern plains of Nepal called the 'terai'. The altitude of the terai is approximately 250 feet above mean sea level. The climate is hot and wet during the summer monsoon (July–September), cool and dry during winter (October–March) and hot and dry from April to June. A beneficial impact of vitamin A supplementation on child mortality has been well established in this area,[8, 9] where xerophthalmia affects 2–3% of preschool children.[10–12] A randomized community trial, to assess the benefit of weekly, maternal, low-dose, vitamin A and β-carotene supplementation on fetal and early infant survival and growth is currently underway in 30 village development committees (VDC) of this district covering a total population of ~165 000 people. Supplements are delivered by local female workers who also identify pregnancies. Women who report being pregnant are enrolled for an initial in-depth interview about diet, morbidity history, and other risk factors in the previous week.

Surveillance of night blindness

A surveillance system for identifying night blind pregnant women was set up in August 1994 as part of the pregnancy surveillance for the above trial in 19 of the 30 study VDC that were accessible by road. A weekly history of XN was obtained by the female workers from the time women reported being pregnant until the termination of pregnancy. Women were asked: 'Are you currently suffering from night blindness?' using local terms for night blindness (*rataunі* in Maithili and *rataundho* in Nepali). Day blindness in women reporting XN was verified by asking them if they could see clearly during the daylight hours. Those reporting XN who had poor or no vision in the evening or at night, but who had no difficulty seeing during the day were defined as night blind cases and were eligible for enrolment in the study.

Study design

A population-based, pair-matched, case-control design was used to examine risk factors associated with XN. Cases and controls (n = 117 pairs), for the present analysis, were restricted to the cohort of pregnant women derived from the placebo arm of those enrolled in the maternal supplementation trial. After the initial five prevalent cases with pre-existing XN identified in the first week of surveillance, the remaining were all incident cases. For each verified case, a control was randomly selected from a list of non-night blind pregnant women from the 19 VDC and matched with the case with respect to months pregnant at the time of enrolment. Three women initially assigned control status, but who subsequently developed XN in pregnancy, were re-designated as cases, randomly assigned a matched control, and interviewed and examined a second time as a case.

After selection, cases and controls were visited in their homes within a week for an interview and examination. Data were collected on women's previous 7-day frequency of intake of certain vitamin A foods and morbidity history, breastfeeding practice, previous history of XN, and a number of household characteristics and socioeconomic status (SES). Capillary blood was drawn by finger stick for serum retinol, β-carotene, and haemoglobin (Hb) estimation. Anthropometric assessment included weight, height, mid upper arm circumference (MUAC), and triceps skinfold (TSF), using the methodology described by Gibson.[13] Arm muscle and fat area were calculated using the formulas given by Gibson.[13] Wasting status was assessed by calculation of the body mass index (BMI) (wt [kg]/ht^2[m]). Blood was collected in microtainers and stored in ice in the field, centrifuged within 4–5 h of collection, and the serum obtained stored and transported in liquid nitrogen to the Center for Human Nutrition Laboratory of the Johns Hopkins University in Baltimore, USA. Once in Baltimore, serum samples were stored in a −70°C freezer until the time of analysis. Retinol and β-carotene were analysed by the reversed phase HPLC method. Haemoglobin concentration was measured using a portable, photometric haemoglobinometer (HemoCue Inc., Mission Viejo, California, USA). Severe anaemia in women was treated with 30 days supplement of iron-folate. Serum retinol and β-carotene values were available for 108 pairs; missing values resulted either due to refusal to give blood or insufficient quantities of serum collected and this occurred randomly in cases and controls.

Statistical analysis

All analyses were performed in SAS version 6.1 (SAS Institute Inc., Cary, North Carolina, USA). Frequency distributions, stem and leaf plots, histograms and scatter plots were used to examine the central tendencies, ranges and distributions of all continuous variables such as age, serum retinol levels, anthropometric measurements and Hb levels. Frequency distributions of categorical and ordinal variables were also examined. This was an aid in deciding appropriate cutoffs to use for presenting descriptive data. Contingency tables were prepared in order to make comparisons between cases and controls by SES factors, weekly food intake, weekly morbidity history and other risk factors. To examine univariate differences in dichotomous variables between

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cases and controls, McNemar's odds ratios (OR) using the ratio of discordant pairs and test based, 95% confidence intervals (CI) were calculated.[14] The paired t-test was used to compare the mean differences in continuous variables between cases and controls.

Multivariate conditional logistic regression analysis that calculated maximum likelihood estimates[14] was done to examine the associations between risk factors and case-control status. Case status was the dependent variable, while main effects examined were vitamin A food intake, serum retinol, anthropometric and Hb measurements, and past 7-day morbidity history, controlling for the confounding effects of some variables like MUAC, and SES. The OR estimates were calculated from the logistic regression parameter estimates and 95% CI from their standard errors by published methods.[14] The independent variables were entered into the models as dichotomous or indicator variables when they were categorical.

Guttman scaling of socioeconomic status

Guttman scaling, also known as scalogram analysis or cumulative scaling, was used to develop an SES scale using household SES characteristics.[15] This procedure is designed to order both items and subjects with respect to some underlying cumulative dimension. Guttman analysis was done with the SAS version 5.1 (SAS Inc.) PROC Guttman procedure using various combinations of dichotomized SES variables. After several iterations of this procedure, using various possible combinations of SES items, the combination of variables that produced the best scale was ownership of (a) land (b) cattle (c) upper storey to the house and (d) cart. A five-point scale that ranged from 0 to 4, was computed by summing the four SES dichotomous variables. This scale was further dichotomized into <2 versus ≥2 in order to classify families into low and high SES groups. This dichotomous variable correlated well with other indicators such as literacy and caste and was used in the multivariate analysis to control for SES.

Ethical approval

Approval to conduct the study was obtained from the Nepal Health Research Council, Kathmandu Nepal, and the Committee on Human Research at the Johns Hopkins School of Hygiene and Public Health, in Baltimore, MD, USA. Verbal consent was obtained from each subject before enrolling them in the study.

Results

Cases and controls were similar with respect to age and the number of previous pregnancies but varied by SES (Table 1). Night blind women were less likely to be literate than controls, (OR = 0.47, not statistically significant), belonged to the low Hindu-Vaishya or Shudra castes (OR = 4.0), were less likely to own large plots of land (OR = 0.31), have an upper storey to their house (OR = 0.44), and were twice as likely to have homes made from mud and thatched walls or roofs than their controls. Using the Guttman scale case households were half as likely as control households to be of higher SES.

The mean gestational age of both cases and controls at the time of the home-based assessment was 7 months. Cases were 4.5 times more likely (95% CI : 2.2–9.0) to report XN in a previous pregnancy than controls. Both groups were equally likely to breastfeed during pregnancy; about 70% of the multiparous women had breastfed their previous child into the current pregnancy.

Table 1 Demographic and socioeconomic characteristics of cases and controls

	Case (%)	Control (%)	Matched odds ratio	95% CI
Age (years)				
<20	13.8	18.8	1.0	–
20–29	68.9	65.8	1.56	(0.68–3.57)
≥30	17.2	15.4	1 33	(0.3–5.93)
Previous pregnancies				
≤2	55.2	53.0	1.0	–
>2	44.8	47.1	1.08	(0.62–1.89)
Literate	5.9	12.6	0.47	(0.19–1.12)
Caste[a]				
Hindu-High	11.9	26.8	1.0	–
Hindu-Low	69.3	50.5	4.0[b]	(1.62–9.89)
Other	18.8	22.7	1.50	(0.25–8.87)
Land				
1–9 kattha[c]	34.8	32.9	0.88	(0.32–2.41)
≥10 kattha	30.3	42.8	0.31[b]	(0.15–0.65)
Thatch/mud wall	68.1	45.3	2.47[b]	(1.48–4.14)
Thatch roofed	41.8	30.2	1.83[b]	(1.04–3.23)
Upper floor	32.5	49.6	0.44[b]	(0.24–0.80)
High SES scale[d]	56.4	73.9	0.47[b]	(0.27–0.82)

[a] High: Brahmin or Chetri, Low: Vaishya or Shudra, Other: Muslim and Buddhist.

[b] Identifies odds ratios with 95% CI that exclude 1.0.

[c] One kattha = 0.034 hectare or 0 084 acre

[d] Dichotomized Guttman scale (<2 versus ≥2) ranging from 0 to 4 created using ownership of land, cattle, upper floor and cart.

The mean serum retinol level of cases was significantly lower (by ~0.30 μmol/l [1 μg/dl = 0.035 μmol/l]) than of controls; cases were four times more likely to have serum retinol levels <0.70 μmol/l and almost three times more likely to have serum retinol levels <1.05 μmol/l (Table 2). A single case who had treated herself with vitamin A and whose serum retinol was 3.09 μmol/l was excluded from this analysis. Mean β-carotene levels were extremely low in both cases (0.22 μmol/l) and controls (0.24 μmol/l) and did not vary significantly by case-status. A large proportion of women in both groups (46% of cases and 26% of controls) had β-carotene levels that were undetectable (<0.035 μmol/l). However, cases were three times more likely to have undetectable β-carotene levels than controls.

Women with XN were more likely to have low anthropometric status reflected by lower body weight (–2.57 kg), BMI (–0.79), MUAC (–0.88 cm), and constituent upper arm muscle (–182 mm^2) and fat (–114 mm^2) areas (Table 3). While cases were 1.3 cm shorter in height than controls, this difference was not statistically significant. Haemoglobin levels among cases were lower, on average by 0.67 g/dl, and three times more likely to be <7 g/dl (representing severe anaemia) than controls (95% CI : 1.25–7.23).

Cases were 2–3 times less likely to have consumed both preformed and carotenoid sources of vitamin A in the past week

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Table 2 Serum retinol and β-carotene levels of cases and controls

	Case (n = 85) Mean (SD)	Control (n = 90) Mean (SD)	Mean diff.[a] (n = 107 pairs)	SE	*P*-value[b]
Retinol (μmol/l)	0.72 (0.41)	1.03 (0.39)	−0.30	0.05	<0.001
β-carotene (μmol/l)	0.22 (0.37)	0 24 (0.37)	−0.02	0.04	NS
	Case (%)	**Control (%)**	**Matched odds ratio**		**95% CI**
Serum retinol (<0.7 μmol/l)	51.8	21.0	4.0[c]		(2 17–7.37)
(<1.05 μmol/l)	76.8	56.1	2.5[c]		(1.37–4.55)
β-carotene (undetectable)	46.0	26.3	3.3[c]		(1.65–6.72)

[a] Case-control.

[b] Using paired t-test.

[c] Identifies odds ratios with 95% CI that exclude 1.0

Table 3 Matched mean difference in anthropometric and haemoglobin measurements between cases and controls

	Case Mean (SD)	Control Mean (SD)	Mean Diff.[a]	SE	*P*-value[b]
Weight (kg)	45.2 (5.2)	47.7 (5.0)	−2.57	0.63	<0.001
Height (cm)	149.6 (5.2)	151.0 (5.9)	−1.30	0.72	<0.08
Body mass index (wt/ht^2)	20.2 (1.9)	20.9 (1.8)	−0.79	0.22	<0.001
Mid upper arm circumference (cm)	21.8 (2.1)	22.7 (1.6)	−0.88	0.23	<0.001
Triceps skinfold (mm)	8.7 (2.8)	9.5 (2.6)	−0.84	0 34	<0 02
Muscle area (mm^2)	2919 (479)	3102 (412)	−182	56	<0.002
Fat area (mm^2)	902 (346)	1014 (300)	−114	40	< 0.005
Haemoglobin (g/dl)	8.9 (2 1)	9.6 (1.6)	−0.67	0.22	<0.004

[a] Case-control

[b] Using paired t-test.

than controls (Figure 1). The relationship was apparent throughout the year, but strongest and statistically significant only in the summer/monsoon months of April–September. In the winter months of October–March, cases and controls were equally likely to have consumed meat and fish sources of vitamin A but in the summer, cases were less likely to eat these foods than controls. Milk and milk products (whey and cream) and provitamin A sources (DGLV and mango) were protective year-round but significantly so only in summer. The protective effect was mainly driven by milk and DGLV consumption.

Seven-day morbidity histories obtained at the time of the XN episode in cases (but during a corresponding gestational time in controls) showed that cases were 2–3 times more likely to report symptoms of urinary/reproductive tract infections that included lower abdominal pain, vaginal discharge and painful and burning urination, diarrhoea/dysentery, and symptoms of nausea, vomiting or poor appetite than controls (Table 4). These symptoms were reported more frequently by cases than controls earlier in pregnancy; ~10 weeks prior to cases developing XN. Women who went on to become cases were also more likely to have reported symptoms of eclampsia (convulsions) or pre-eclampsia (swelling of hands or face) in the previous week (OR = 2.51, 95% CI : 1.13–5.67).

Multivariate conditional logistic regression analysis that included in the model all four classes of risk factors associated with XN, namely diet (weekly vitamin A food intake, using milk and dark green leaves—the two foods that had the most consistent protective effect against XN), protein-energy nutritional status (MUAC), anaemia (Hb), and morbidity (positive weekly history of lower abdominal pain, diarrhoea/dysentery or nausea) as independent variables, showed that OR associated with each class of factors were significant after adjusting for each others'

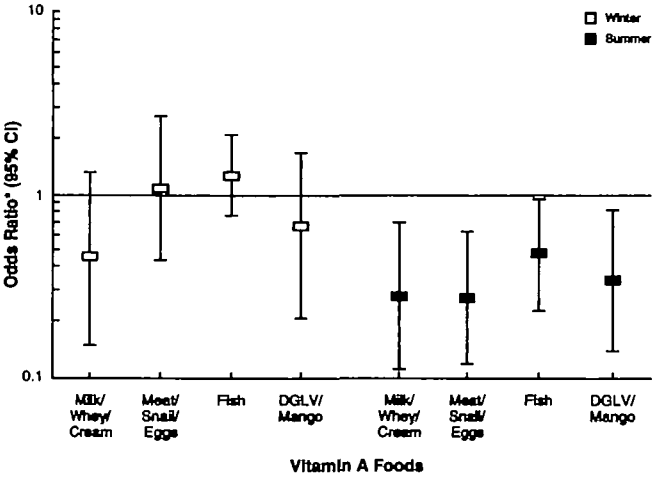

Figure 1 Odds of consuming combinations of foods rich in vitamin A in the past week by season among cases of maternal night blindness relative to non-night blind pregnant controls

* Odds ratios adjusted for mid upper arm circumference and socioeconomic scale using conditional logistic regression.

DGLV—dark green leafy vegetables.

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Table 4 Prevalence of morbidity symptoms in the previous week among cases and controls

	During night blindness			Prior to night blindness[a]		
Morbidity	Case (%)	Control(%)	OR[b]	Case (%)	Control (%)	OR[b]
Urinary/reproductive tract infections[c]	64.6	44.4	2.14[d]	42.6	25.6	1.90[d]
Diarrhoea/dysentery	20	5.2	3.45[d]	5.2	5.9	0.57
Pre-eclampsia/eclampsia[e]	39.1	29.7	1.66	27.8	17.1	2.51[d]
Nausea/ vomiting/ poor appetite	48.2	28.7	2.10[d]	33.9	19.7	2.67[d]

[a] Morbidity history collected ~10 weeks prior to cases developing night blindness.

[b] Odds ratios (OR) adjusted for mid upper arm circumference, socioeconomic scale and age of the woman using conditional logistic regression.

[c] Any lower abdominal pain, painful urination or vaginal discharge.

[d] Identifies OR with 95% CI that exclude 1.0

[e] Any convulsions or swelling of hands/face

Table 5 Relative contribution of weekly diet (vitamin A food intake), acute protein-energy malnutrition (mid upper arm circumference [MUAC]), anaemia (haemoglobin [Hb]) and morbidity as risk factors for night blindness during pregnancy

Independent variables	Odds ratio[a]	95% CI
Vitamin A-foods[b] (1–3×)	0.56	(0.24–1.31)
(≥4×)	0.19[c]	(0.08–0.49)
Morbidity[d]	3.28[c]	(1.69–6.34)
MUAC (cm)	0.77[c]	(0.63–0.94)
Hb (g/dl)	0.79[c]	(0.64–0.97)
Socioeconomic scale (≥2)	0.66	(0.32–1.34)

[a] Adjusted using conditional logistic regression

[b] Vitamin A foods: frequency of milk or dark green leaves consumption in the past week

[c] Identifies odds ratios with 95% CI that exclude 1 0.

[d] Morbidity: any lower abdnominal pain, diarrhoea/dysentery or nausea in the past week during XN episode in cases

effects (Table 5). Removing any one of the variables from the model left virtually unchanged the OR estimates for the other covariates, indicating that all four risk factors were independently and strongly associated with XN in pregnant women.

Discussion

Night blind pregnant women were vitamin A-deficient, having, on average, a serum retinol concentration ~33% (0.30 μmol/l) lower than non-night blind controls of similar gestational age. Half of the cases had serum retinol levels <0.70 μmol/l compared to only one-fifth of controls. On matched-analysis, cases had a fourfold higher risk of hyporetinaemia than controls. Although serum β-carotene levels were exceedingly low in both groups of women, the odds of an undetectable level was threefold higher in cases relative to controls. A heightened risk of vitamin A deficiency with XN is fully consistent with experimental data showing vitamin A depletion to cause retinal rod cell dysfunction in animals,[16] impaired dark adaptation in adults,[17,18] and clinical studies in children showing a high risk of hyporetinaemia with XN, and rapid and definitive cure or effective prevention with vitamin A supplements.[19] However, excessive risk of XN persisted in pregnant women at serum retinol levels of 0.70–1.05 μmol/l, suggesting that symptoms can arise in the presence of physiologically significant and acceptable levels of circulating retinol. This suggests that the standard cutoff of <0.70 μmol/l specified by the World Health Organization, as applied typically to children,[20] may be inappropriate for pregnant women. Notably, classic, controlled depletion studies in adults revealed impaired dark adaptation at serum vitamin A levels of 0.70–1.05 μmol/l[17,18] and, in Indonesia, ~20% of vitamin A-responsive night blind children had initial serum retinol levels >0.70 μmol/l.[21]

The relatively late gestational appearance of XN (~7 months) is consistent with the small additional requirement for vitamin A by women confined mostly to the third trimester.[22] Maternal plasma retinol typically decreases during pregnancy[23–27] due to plasma volume expansion. However, in poor, undernourished populations this decline may reflect more than normal haemodilution. Women habitually consuming basal amounts or less of vitamin A are likely to have very poor reserves, and in the absence of dietary improvement during pregnancy, may fail to meet the additional requirements of pregnancy, particularly during the third trimester. Physiological changes combined with a high risk of infection could precipitate acute vitamin A deficiency and lead to XN late in pregnancy.

Other conditions, however, may increase the risk of gestational XN. Protein-energy malnutrition (PEM) could contribute to XN by reducing retinol-binding protein synthesis and thereby decreasing the transport of retinol from the liver into circulation.[28,29] It could also lower synthesis of opsin—the protein moiety of the rod photoreceptor pigment (rhodopsin) in the retina—which could lead to impaired dark adaptation.[16] Night blind women in Nepal were more severely malnourished than their controls by all anthropometric indicators. Elsewhere, low protein-energy status has been associated with hyporetinaemia in pregnant Indonesian,[30] Nigerian[31] and Navajo[32] women, though not to a level that manifested XN. In India, mildly xerophthalmic pregnant women had lower mean weight (43 versus 51 kg) and height (146 versus 153 cm) than non-xerophthalmic ones.[33] A deficiency in zinc, which is required for dark adaptation,[34,35] could also explain the occurrence of XN in the presence of adequate vitamin A status in populations such as rural Nepal, where intakes of meat and fish (sources of zinc) are low and dietary levels of inhibitors of zinc absorption (e.g. phytates from grains) are high.[36] Interim findings of the large trial in Nepal suggest incomplete efficacy of weekly, low-dose vitamin A supplementation in preventing XN in pregnancy (Christian *et al.*, unpublished data).

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Night blind women consumed vitamin A-rich foods (e.g. milk, snail, fish, dark green leaves, etc.) less frequently than controls late in pregnancy when XN was apparent, especially in the premonsoon and monsoon 'summer' (April–September). Preliminary observations suggest that the occurrence of XN is highest during the summer months, when the number of pregnant women in the last trimester also peaks. Lower availability and higher prices of some of these foods during this time of year may partly explain the seasonally decreased intake.

Cases were at higher risk of anaemia than controls, supporting previous evidence that vitamin A deficiency contributes to low haemoglobin levels.[19] Xerophthalmic pregnant women in India had lower haemoglobin levels (8.3 g/dl) than their non-deficient peers (10.8 g/dl),[33] similar to the differences observed in our study (8.9 versus 9.6 g/dl). The association could be indirect, paralleling a poorer overall nutritional status of night blind women, but could also be causal, illustrated by a significant haemoglobin response to vitamin A supplementation observed in pregnant Indonesian women.[37]

Night blind women were more susceptible to illness, both infectious (e.g. symptoms of urinary and possibly reproductive tract infections, diarrhoea and dysentery) and non-infectious (e.g. symptoms of pre-eclampsia or eclampsia, and poor appetite, nausea and vomiting). Cases were also at greater risk than controls of morbidity (except for diarrhoea/dysentery) when they were assessed 2–3 months earlier in pregnancy; a time that preceded their episode of XN. This estimate of antecedent risk lacks any bias that otherwise could attend morbidity ascertainment at the time cases were night blind. It, thus, could represent effects of subclinical vitamin A deficiency, other predisposing nutritional deficiencies and health risks, as well as, possible (bi-directional) effects of early pregnancy morbidity on later vitamin A status. Vitamin A deficiency has been a long established risk factor for infection, and *vice versa*, especially with respect to persistent diarrhoea and measles in children[19] but evidence of this association in pregnant women has been scant except among HIV-infected subjects.[38,39] Pre-eclampsia or toxaemia in pregnancy has been occasionally associated with vitamin A deficiency and night blindness.[1,26,40] Stronger evidence has emerged in recent years linking low circulating levels of β-carotene,[41,42] a known antioxidant, and serum vitamin A[42] to pre-eclampsia. Infection has only infrequently been associated with vitamin A deficiency in women. Bacteriuria, suggestive of urinary tract infection, was observed in ~80% of pregnant women presenting with nyctalohemeralopia.[43] Lowered vitamin A status during pregnancy has also been shown to increase susceptibility to reproductive tract infection in the puerperium in US women,[44] while a higher vitamin A intake has been linked to a decreased risk of diarrhoea in Bangladeshi women.[45]

We conclude that night blind pregnant women in rural South Asia are exposed to an excess risk of vitamin A deficiency, a known proximal determinant of this condition. In addition, and independent of their vitamin A status, night blind women may be anaemic, acutely malnourished with respect to protein, energy and (likely) other micronutrients such as zinc, and prone to increased risk of infectious and non-infectious illnesses throughout pregnancy. To our knowledge, this is the first population-based report of increased risk of morbidity associated with XN during pregnancy. A low social standing and poverty appear to contribute to this multifaceted syndrome. However, a simple history of maternal XN may help to identify high-risk pregnancies and should be included in antenatal assessment and care programmes on the Indian subcontinent. Based on this evidence, treatment for XN in pregnancy should include vitamin A (although treatment regimens have not been established and efficacy may only be partial). In addition, night blind women may require additional nutritional and antenatal care not previously considered.

Acknowledgements

We thank members of the Nepal Nutrition Intervention Project —Sarlahi in Nepal (beyond the authors) for their assistance in the conduct of this study, especially DN Mandal, TR Shakiya, G Subedi, U Shankar, A Bhetwal, DB Khadka, SV Shrestha, K Dhakal, I Shrestha, and Drs Margaret E Bentley and Rebecca J Stoltzfus for their input and discussion of the results. The study was funded through Cooperative Agreement No DAN 0045-A-00–5094–00 between the Center for Human Nutrition, Department of International Health and the Dana Center for Preventive Ophthalmology, the Johns Hopkins University, Baltimore, MD, USA and the Office of Health and Nutrition, US Agency for International Development, Washington DC, with additional support from Task Force Sight and Life (Roche, Basel, Switzerland).

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